U.S. Department of Homeland Security

Science and Technology Directorate

Resilient Precision Time Pilot

Statement of Work

I. PURPOSE

(TBD)

II. OBJECTIVE

The objective of this effort is to develop and demonstrate a pilot that is reproducible, scaleable, and sustainable for use by end-users, and is resilient to Positioning, Navigation, and Timing (PNT) threats. The resulting products from this work shall be a reproducible and enduring pilot capability that can be replicaded and scaled in other states.

III. BACKGROUND

Positioning, Navigation, and Timing (PNT) services have become an invisible but essential utility for many critical infrastructure operations. Executive Order 13905, "Strengthening National Resilience Through Responsible Use of Positioning, Navigation, and Timing Services," indicates that the "the disruption or manipulation of these [PNT] services has the potential to adversely affect the national and economic security of the United States."

Precision timing from PNT services are particularly essential to many critical infrastructure sectors, and this precision timing is predominantly provisioned through the Global Positioning System (GPS) and other Global Navigation Satellite Systems (GNSS). However, GPS and GNSS are vulnerabile to intentional and unintentional interference and the frequency of some incidents has become increasingly frequent.

To address this strategic risk to U.S. critical infrastructure, this project will pilot an effort to (1) establish a regional network of resilient timing nodes and (2) a means of distributing precision timing to critical infrastructure operators in the local area. While this pilot will be conducted in the state of Texas, the products of this effort shall be scalable and reproducible to allow other states to replicate a similar capability to enable resilient operations for their critical infrastructure.

This effort shall leverage other excisting commercial timing standards, government publications, and resources on PNT resilience where applicable, such as the DHS Resilient PNT Reference Architecture, DHS Resilient PNT Conformance Framework, DHS PNT Best Practices, and the NIST PNT Profiles.

The total estimated cost of this (TBD).

The following documents may be helpful to the Contractor in performing the work described in this document:

DHS Resilient PNT Reference Architecture, V1.0

https://www.dhs.gov/science-and-technology/publication/resilient-pnt-reference-architecture

DHS Resilient PNT Conformance Framework (Version 2.0)
https://www.dhs.gov/publication/st-resilient-pnt-conformance-framework

DHS PNT Best Practices (September 2024)

https://www.dhs.gov/science-and-technology/publication/pnt-best-practices-supporting-critical-infrastructure

DHS GPS Best Practices (January 2017)

https://www.cisa.gov/sites/default/files/documents/Improving_the_Operation_and_Development_of_Global_Positioning_System_%28GPS%29_Equipment_Used_by_Critical_Infrastructure_S508C.pdf

NIST PNT Profiles (Latest Version)

https://csrc.nist.gov/pubs/ir/8323/r1/final

Other Resources on GPS.gov

https://www.gps.gov/resilience-through-responsible-use-pnt

Definition of Critical Infrastructure Sectors

https://www.cisa.gov/topics/critical-infrastructure-security-and-resilience/critical-infrastructure-sectors

IV. SCOPE OF WORK

The Contractor shall develop and demonstrate a pilot demonstration of a resilient regional PNT timing network architecture for use by critical infrastructure end-users. This pilot shall consist of (1) a master timing node network; (2) traceability to UTC(NIST); and (3) “last mile” timing distribution to critical infrastructure end-users.

The Contractor shall work with stakeholder partners for all three portions of the pilot. Additionally, the Contractor shall develop documentation and other necessary resources to enable other states to reproduce a similar capability.

Figure 1 shows a conceptual architecture diagram for the pilot effort. The master timing node network consists of Primary Nodes, Secondary Nodes, and Distribution Nodes. The Primary and Secondary nodes shall contain high quality holdover clocks that can continue to operate for a period of time without external timing source input, such as PTP and/or GPS. The Primary Node will have higher quality clocks than the Secondary Node, which also increases its cost. These nodes shall also incorporate concepts from the DHS Resilient PNT Reference Architecture to ensure PNT resilience and treat all external PNT sources as attack surfaces.

The initial pilot may contain a downscaled subset of the timing node network due to funding constraints. However, the pilot must still demonstrate the overall architecture while ensure it is reproducible and provides usable timing information to end-users.

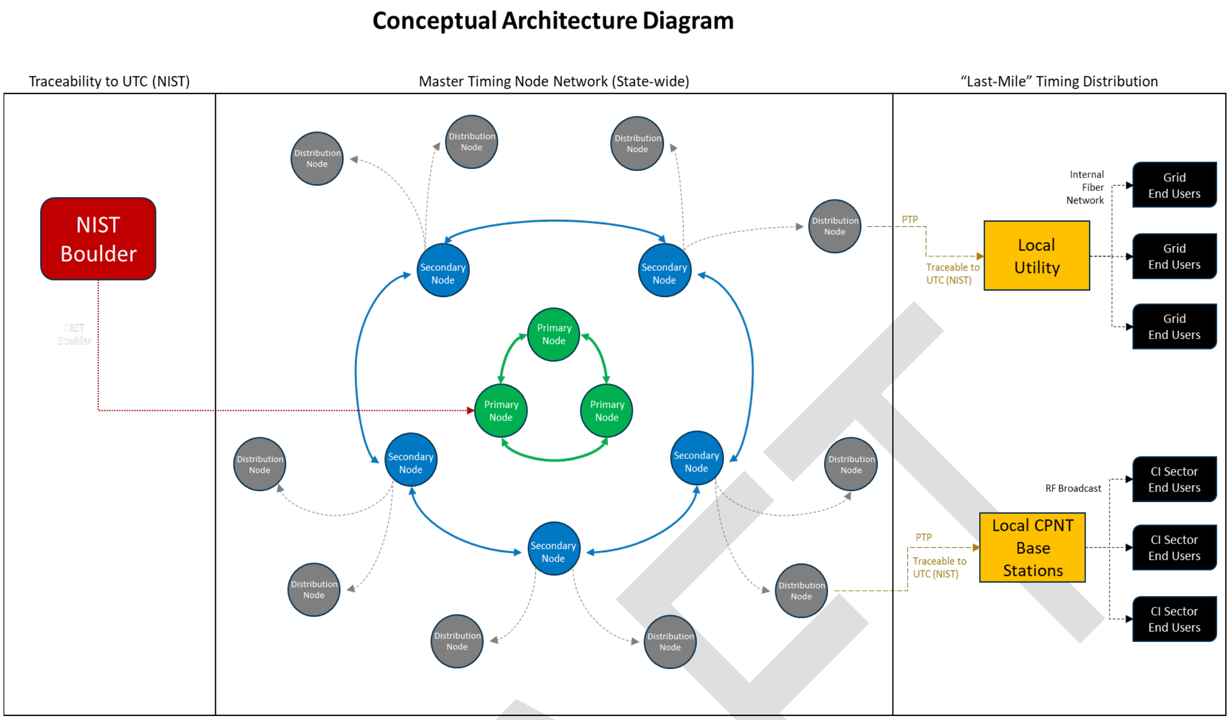

Figure 1. Conceptual Architecture Diagram (see Appendix A for larger version)

Specific Requirements / Tasks

The pilot must meet the following minimum threshold requirements:

- Master Timing Node Network
  - Accuracy: 100 nanoseconds (or better)
  - Holdover (Primary Node): 30 days
  - Holdover (Secondary Node): 1 day (Threshold) / 5 days (Objective)
- "Last Mile" Timing Distribution to End-User
  - Accuracy Threshold: 1 microsecond
  - Accuracy Objective: 500 nanoseconds or better
- Timing at both levels must be traceable to UTC(NIST)

TASK 1.0: Project Management

Manage cost, schedule, and performance of the project and meet with DHS S&T monthly to provide updates on project progress. Provide a Research Management Plan (RMP) within 30 calendar days of contract award. This plan outlines how the contract manages the execution of all tasks to include project planning, scheduling, monitoring project progress, risk management, and cost management. Additional project management activities include:

- **Kick-Off Meeting:** Complete a post-award kick-off review with customer, to be held within 30 calendar days of the contract award.
- **Data and Records Management:** Manage the collection, storage, reporting, and archiving of all necessary data monthly.
- **Development Approach:** Identify internal/external dependencies and interdependencies, and backup options for implementation.
- **Risk Mitigation and Planning:** Document programmatic risks and opportunities with appropriate assessment and analysis. Ensure production and maintenance of all necessary data to adequately document risk decisions.
- **Monthly Status Report:** Submission of a monthly report incorporating a quad chart format (template provided at Kick-Off meeting), compiling the following information from the previous month:
  - » Project activities
  - » Deliverables table with planned, revised, and actual dates in two-digit month/day/year format
  - » Risks and mitigations, problems encountered, and recovery plans (when necessary)
  - » Significant meetings attended
  - » Preparation for upcoming scheduled activities
  - » Expenditure report showing current expenditure outlined by task with charges to each labor category, as well as past, cumulative, and anticipated expenditures
- **Quarterly Project Review:** Preparation and execution of a Biannual Project Review twice a year, with briefing materials provided in advance of meeting date. The purpose of the biannual project review is to present progress and findings to date and ensure all milestones and deliverables are on track as described in this TORP.
- **Final Report:** Completion of a final report 30 calendar days prior to the end of the period of research performance. The report shall analyze the project requirements, the proposed solution and curricula, the actual curricula, and all outcomes and other knowledge products developed under the contract award.

TASK 2.0: Architecture Design Assessment

The Contractor shall identify and develop a set of options for both the master timing node network and local distribution methods to critical infrastructure end-users.

The design assessment shall include the full set of options and describe general considerations for each, which will be important for others that seek to replicate this architecture. The assessment shall also evaluate the viability of each option for this particular pilot effort, along with generalized considerations that can be used for others when replicating this architecture.

Selection of implemented options for the pilot will occur in Task 4.0 with DHS sponsor involvement.

2.1: Deliverable: Draft Architecture Assessment Document

2.2: Milestone: Architecture Assessment Review

2.3: Deliverable: Final Design Assessment Document

TASK 3.0: Stakeholder Partner Engagement & Collaboration

The Contractor shall identify, document, and coordinate partnerships with stakeholders necessary to successfully pilot this effort. This includes partners both for providing UTC timing and traceability to the master timing node network as well as the "last mile" timing distribution to local critical infrastructure end-users.

For example, providing UTC(NIST) timing to the master timing network could involve partner with the National Institute of Standards and Technology (NIST), fiber optical communication carriers for long-haul fiber connections, or Internet2 consortium members for access to their fiber optic backbone network. Examples of critical infrastructure end-users include local power utility companies, telecommunication carriers utilizing precision timing for their local equipment, or local emergency service sector operators.

3.1: Milestone: Coordinate Partnerships for UTC(NIST) Traceability

3.2: Milestone: Coordinate Partnerships for Master Timing Node Network

3.3: Milestone: Coordinate Partnerships for "Last Mile" Timing Distribution

3.4: Deliverable: Stakeholder Partner Plan

TASK 4.0: Pilot Architecture

The Contractor shall develop a architecture implementation design specific to this initial pilot for the state of Texas. This design shall be based upon input from the DHS sponsor as well as how the information and constraints from both Task 2.0 (Design Assessment) and Task 3.0 (Stakeholder Partner Engagement & Collaboration) apply to the situation for this particular pilot.

The selection of options from Task 2.2 is expected to be highly dependent on available connectivity and infrastructure from stakeholder partners. Implementation cost is also expected to be a significant factor. The DHS sponsor will provide input for tradespace considerations, which affect the likelihood of success for this effort.

Even if constraints cause the implementation to be scaled down in scope, the chosen options for the implementation shall be representative enough to demonstrate the architecture and its reproducibility and usability.

4.0.1: Deliverable: Draft Pilot Implementation Design

4.0.2: Milestone: Design Review

4.0.3: Deliverable: Final Pilot Implementation Design

TASK 4.1: Pilot Architecture Development

The Contractor shall develop, build, and integrate a pilot architecture based upon the implementation selections made in Task 4.0 (Pilot Architecture). This shall include the master timing node network, “last mile” distribution, and integration with stakeholder partners on both. The developed pilot shall meet the requirements specified in earlier in Specific Requirements / Tasks.

The developed pilot architecture, while smaller in scale than the full buildout, must be representative enough to demonstrate the overall architecture and its reproducibility and usability.

After completion of integration, the Contractor shall conduct tests to ensure all requirements for the pilot are met and provide a test report. The report shall contain the test results, key data, data collection methodology, and other relevant details.

4.1.1: Milestone: Completion of pilot timing node network

4.1.2: Milestone: Completion of “last mile” distribution systems

4.1.3: Milestone: Completion of internal and external integration

4.1.4: Deliverable: Test Report

TASK 4.2: Pilot Demonstration

The Contractor shall plan and execute a pilot demonstrate with a VIP day. The audience shall include not only DHS and partner stakeholders but also external parties that may be interested in either building out similar capabilities elsewhere or other potential users. Expected external parties is the Texas State Government (particularly the Texas State Legislature), U.S. Senators for Texas, and representatives from other states.

The objectives for the pilot demonstrate include the following: (1) to raise awareness of the pilot capability; (2) to demonstrate the feasibility and scalability of such a capability; and (3) to inform the Texas legislative session in 2027 to potentially adopt the pilot capability and build it out to full scale for the state’s own use.

Due to the sensitivities and complexities involved, the planning for the pilot demonstration and VIP day shall be performed in collaboration with the DHS sponsor. The DHS sponsor will provide inputs for the planning and the Contractor shall draft a pilot demonstration plan.

The plan shall address the following, at a minimum:

- Schedule: Schedule both before the demonstration and the schedule during execution of the demonstration.
- Demonstration Format: How the demonstration will be conducted. The demonstration can also have multiple segments to address different concerns attendees would be interested in, such as technical, economic, performance, impact, and resilience.
- Demonstration Materials: Identify the materials that must be prepared during the demonstration. This includes presentation materials and pre-recorded data.
- Plan for Live Demonstration of Impact and Resilience: The pilot shall include a live demonstration segment that demonstrates its operations in a simulated GPS outage

event and the resilience it provides to connected critical infrastructure end-users by continuing to deliver precision timing to critical infrastuructre end-users at a performance level that allows continued operations.

- Plan for Pre-Recorded Demonstration Data: Demonstrating resilience in a simulated GPS outage may require collecting data over a period of time that is longer the duration of the VIP day. Therefore the VIP day may also involve presentation of pre-recorded GPS outage data.
- Plan for Dry-run: In coordination with the government sponsor, a dry-run shall be conducted to minimize issues occurring when executing the VIP day.

After execution of the demonstration and VIP day, the Contractor shall deliver a post-event report highlighting the technical results of the demonstration as well as a summary of key takeaway comments and inputs from attendees. The purpose of this report is to support follow-on actions and potential interactions with the Texas State Legislature for future build-out of the pilot capability. The report shall also capture impact metrics to support internal DHS sponsor Return-on-Investment (ROI) assessments.

4.2.1: Deliverable: Task plan for the pilot demonstration and VIP day

4.2.2: Milestone: Completion of demonstration materials

4.2.3: Milestone: Execution of demonstration and VIP day

4.2.4: Deliverable: Post-event report to support follow-on activities

TASK 5.0: Final Report, Transition and Implementation

The Contractor shall develop a transition and implementation package containing documentation, design diagrams, architecture diagrams, design specifications, hardware and software options, architecture design options, and any other resources needed for others to reproduce this project's capability from selection, design, to implementation.

This package shall also contain documentation for all architecture design options and their tradeoffs and considerations, as informed by Task 2.0 (Architecture Design). Implementation options shall included recommendations of hardware and software from a range of different vendors.

5.0.1: Deliverable: Transition and Implementation Package

5.0.1: Deliverable: Final Report

V. SEVERABILITY

The requirement for this project is non-severable. The full benefit for the work performed is not fully realized until all deliverables are submitted and the work is complete.

VI. KEY MILESTONES AND DELIVERABLES

Task Reference	Milestones/Deliverables	Due Date or Completion Date
1.0: Project Management		
	1.1: Deliverable: Draft Project Plan	30 days after award.
	1.2: Milestone: Project Plan Review	45 days after award
	1.3: Deliverable: Monthly Status Reports	Monthly
2.0: Architecture Design Assessment		
	2.1 Deliverable: Final Design Assessment Document	3 months after award
	2.2 Milestone: Design Assessment Review	3 months after award
	2.3 Deliverable: Final Design Assessment Document	4 months after award
3.0: Stakeholder Partner Engagement & Collaboration		
	3.1 Milestone: Coordinate Partnerships for UTC(NIST) Traceability	3 months after award
	3.2 Milestone: Coordinate Partnerships for Master Timing Node Network	3 months after award
	3.3 Milestone: Coordinate Partnerships for “Last Mile” Timing Distribution	3 months after award
	3.4 Deliverable: Stakeholder Partner Plan	4 months after award
4.0: Pilot Architecture Implementation Design		
	4.0.1 Deliverable: Draft Pilot Implementation Design	4 months after award
	4.0.2 Milestone: Design Review	4 months after award
	4.0.3 Deliverable: Final Pilot Implementation Design	6 months after award
4.1: Pilot Architecture Development		
	4.1.1 Milestone: Completion of pilot timing node network	10 months after award
	4.1.2 Milestone: Completion of “last mile” distribution systems	10 months after award
	4.1.3 Milestone: Completion of internal and external integration	12 months after award
	4.1.4 Deliverable: Test Report	12 months after award
4.2: Pilot Demonstration		
	4.2.1 Deliverable: Task plan for the pilot demonstration and VIP day	12 months after award
	4.2.2 Milestone: Completion of demonstration materials	14 months after award
	4.2.3 Milestone: Execution of demonstration and VIP day	16 months after award
	4.2.4 Deliverable: Post-event report to support follow-on activities	18 months after award
5.0: Final Report, Transition and Implementation		
	5.0.1 Deliverable: Transition and Implementation Package	20 months after award
	5.0.1 Deliverable: Final Report	24 months after award

Table #1 Key Milestones and Deliverables

VII. <u>Project Timeline</u>

A summary of all Tasks and scheduled meetings throughout the performance period of the contract appears below. The durations of all Tasks shown as horizontal bars indicate the month(s) during which activities will occur to complete the respective Task.

Indicate month deliverable is due after award or services are completed												
Month	1	2	3	4	5	6	7	8	9	10	11	12
TASK 1.0: Project Management												
1.1: Draft Project Plan	X											
1.2: Project Plan Review	X	X	X	X	X	X	X	X	X	X	X	X
1.3: Monthly Status Reports				X				X				X
TASK 2.0: Architecture Design Assessment												
2.1: Draft Architecture Assessment Document			X									
2.2: Archtecture Assessment Review			X									
2.3: Final Design Assessment Document				X								
TASK 3.0: Stakeholder Partner Engagement & Collaboration												
3.1: Coordinate Partnerships for UTC(NIST) Traceability			X									
3.2: Coordinate Partnerships for Master Timing Node Network			X									
3.3: Coordinate Partnerships for "Last Mile" Timing Distribution			X									
3.4: Deliverable: Stakeholder Partner Plan				X								
TASK 4.0: Pilot Architecture												
4.0.1: Draft Pilot Implementation Design				X								
4.0.2: Design Review				X								
4.0.3: Final Pilot Implementation Design						X						
TASK 4.1: Pilot Architecture Development												
4.1.1: Completion of pilot timing node network										X		
4.1.2: Completion of "last mile" distribution systems										X		
4.1.3: Completion of internal and external integration												X
4.1.4: Test Report												X
TASK 4.2: Pilot Demonstration												
4.2.1: Task plan for the pilot demonstration and VIP day												X
4.2.2: Completion of demonstration materials												
4.2.3: Execution of demonstration and VIP day												
4.2.4: Post-event report to support follow-on activities												

Indicate month deliverable is due after award or services are completed												
Month	13	14	15	16	17	18	19	20	21	22	23	24
TASK 1.0: Project Management												
1.1: Draft Project Plan												
1.2: Project Plan Review	X	X	X	X	X	X	X	X	X	X	X	X
1.3: Monthly Status Reports				X				X				X
TASK 4.2: Pilot Demonstration												
4.2.1: Task plan for the pilot demonstration and VIP day												
4.2.2: Completion of demonstration materials		X										
4.2.3: Execution of demonstration and VIP day				X								
4.2.4: Post-event report to support follow-on activities						X						
TASK 5.0: Final Report, Transition and Implementation												
5.0.1: Transition and Implementation Package								X				
5.0.2: Final Report								X				X

Table #2 Project Timeline

VIII. Other Contract Details

A. PROPOSAL DUE DATE: (TBD)**.**

B. PROPOSAL SUBMISSION REQIREMENTS: The technical and cost proposals should be submitted electronically to the following individuals:

a. **[NAME], Procuring Contracting Officer, [Email]**

b. **[NAME], COR, [Email]**

C. ANTICIPATED AWARD TYPE: (TBD)

D. Anticipated Period of Performance: 24 months from date of award.

E. TORP QUESTIONS: Any questions related to this TORP should also be sent electronically to the Procuring Contracting Officer at the email address identified above.

F. **Place of Performance.** Work performed under this contract shall be conducted at the Contractor's facilities, sites for the Master Timing Node Network, sites for distribution to critical infrastructure end-users, and sites supporting UTC, U.S. Department of Commerce, National Institute for Standards and Technology, Time and Frequency Division, for traceability.

This is anticipated to occur in facilities within the State of Texas and potentially in Boulder, Colorado, where U.S. Department of Commerce, National Institute for Standards and Technology, Time and Frequency Division Boulder is located for UTC traceability.

G. Travel. Travel may be required in the performance of the tasks described herein. No less than 21 days prior to any travel, the contractor shall submit an itemized cost estimate for the trip, covering all proposed attendees, including subcontractors, to the DHS S&T Contracting Officer's Representative (COR). The COR must pre-approve all travel. All travel costs associated with the execution of the tasks indicated in this SOW will be reimbursed in accordance with the limits set forth in the Federal Travel Regulations and the Federal Acquisition Regulations. The DHS S&T COR and the DHS S&T Special Assistant for International Policy must approve any foreign travel in advance.

H. Funding availability. This effort is funded in the amount of (TBD) for a 24-month period of performance.

I. Government-furnished information and property. It is not anticipated that the Government will need to provide information or materials to the contractor in support of tasks under this SOW.

J. Contractor-acquired information and property. Before purchasing any individual item equal to or exceeding $5,000 not included in the contractor's cost proposal that is required to support technical tasks performed pursuant to this contract, the contractor shall obtain the DHS S&T Contracting Officer's prior written consent. If the DHS S&T Contracting Officer consents to such purchase, such item shall become the property of DHS. The contractor will maintain any such items according to currently existing property accountability procedures.

In the administration of Contractor Acquired Property, a listing of all property at or above the $5,000 threshold shall be provided to the DHS S&T Contracting Officer annually from the date of acceptance of the contract. Thirty days prior to the completion of work and acceptance of all deliverables, the final and complete listing of all Contractor Acquired Property at the acquisition cost of $5,000 or more directly charged to this contract shall be provided to the DHS S&T Contracting Officer. The DHS S&T Contracting Officer will determine the final disposition of any such items in writing. The contractor shall not take any action associated with the transfer of the property that will result in an increase of cost to the obligated amount or the total estimated cost unless the DHS S&T Contracting Officer has taken action to increase the obligated amount and total estimated cost prior to the final closeout amendment.

K. Deliverables. The contractor will provide all deliverables identified in this SOW directly to the DHS S&T COR.

L. Invoices. Monthly invoices must be delivered to InvoiceSAT.Consolidation@ice.dhs.gov with cc: to the COR and Contracting Officer listed below, no later than the 15th day of each month.

M. Section 508 Requirements. Section 508 of the Rehabilitation Act, as amended by the Workforce Investment Act of 1998 (P.L. 105-220) (codified at 29 U.S.C. § 794d) requires that when Federal agencies develop, procure, maintain, or use information and communications technology (ICT), it shall be accessible to people with disabilities. Federal employees and members of the public with disabilities must be afforded access to and use of information and data comparable to that of Federal employees and members of the public without disabilities.

All products, platforms and services delivered as part of this work statement that, by definition, are deemed ICT or that contain ICT shall conform to the revised regulatory implementation of Section 508 Standards, which are located at 36 C.F.R. § 1194.1 & Apps. A, C & D, and available at https://www.gpo.gov/fdsys/pkg/CFR-2017-title36-vol3/pdf/CFR-2017-title36-vol3-part1194.pdf. In the revised regulation, ICT replaced the term electronic and information technology (EIT) used in the original 508 standards.

N. Security Requirements. Work under this SOW will not involve access to classified information. If provided DHS “sensitive” information (e.g., items marked with For Official Use Only (FOUO) or other appropriate marking), the contractor agrees it shall safeguard such information by not providing access of this marked information to any non-federal personnel unless advance approval is obtained from the DHS S&T COR. In turn, the DHS S&T COR must that DHS Non-Disclosure Agreement (NDA) Form 11000-6s are signed by non-federal personnel before access to DHS “sensitive” information is given to them. The DHS S&T COR must further obtain copies of the executed, signed DHS Form 11000-6s, to be provided to the DHS Contracting Officer (for inclusion in the official DHS Office of Procurement Operations (OPO) contract file.

DHS has and will exercise full control over granting, denying, withholding, or terminating unescorted Government facility and/or sensitive Government information access for Contractor employees, based upon the results of a background investigation. DHS may, as it deems appropriate, authorize, and make a favorable entry of duty (EOD) decision based on preliminary security checks. The favorable EOD decision would allow the Contactor to commence work temporarily prior to the completion of the full investigation. The granting of a favorable EOD decision shall not be considered as assurance that a full employment contractor fitness (suitability) authorization will follow as a result thereof. The granting of a favorable EOD decision or a full Contractor fitness (suitability) authorization determination shall in no way prevent, preclude, or bar the withdrawal or termination of any such access by DHS, at any time during the term of the award. No employee of the Contractor shall be allowed unescorted access to a Government facility, access to any sensitive information or access to DHS IT Systems without a favorable EOD decision or Contractor fitness (suitability) determination by the DHS Office of Security. Contract employees assigned to the award not needing access to sensitive DHS information or recurring access to DHS facilities will not be subject to security contractor fitness (suitability) screening. Contract employees waiting and EOD decision may not begin work on the award. Limited access to Government buildings is allowable prior to the EOD decision if the contractor is escorted by a Government employee. This limited access is to allow contractors to attend briefings, nonrecurring meetings and begin transition work. Classified information is Government information which requires protection in accordance with Executive Order 13526, National Security Information (NSI) as amended and supplemental directives. If the Contractor has access to classified information at a DHS owned or leased facility, it shall comply with the security requirements of DHS and the facility. If the Contractor is required to have access to classified information at another Government facility, it shall abide by the requirements set forth by the agency.

O. The S&T Privacy Office, Office of the Chief Information Officer, and the Office of the Chief Security Officer require the insertion of the HSAR 15-01 Safeguarding of Sensitive Information (March 2015) and HSAR 15-01 Information Technology Security and Privacy Training (March 2015). See HSAR 15-01.

P. **Information Release.** Prior to releasing any information developed using funds awarded under this contract (or task order, as applicable), the contractor is required to route all materials to the S&T COR who will ensure DHS S&T has approved the content release. A minimum of four (4) weeks must be allowed for the review and screening of such information (including but not limited to articles, presentations, videos, speeches at conferences, pamphlets, and other forms of printed media) to ensure that such information's release does not violate security policies and procedures.